

SUNGENE INDUSTRIAL IOT

Remote Monitoring & Data Acquisition

Remote monitoring solutions for water, energy
and industrial equipment

LoRaWAN . NB-IoT . RS485 . Modbus . 4G LTE

sungeneiot.com | contact@sungeneiot.com

Product Catalog 2026 - Edition v2

Remote monitoring, by scenario

Water Monitoring

Tank level monitoring, water leak detection and pump monitoring for water utilities, industrial facilities and agriculture - using LoRa / NB-IoT wireless sensors and gateways.

Energy Monitoring

Smart metering, energy data collection and solar site monitoring to spot loss, downtime and underperformance in time, across energy, solar and telecom sites.

Equipment Monitoring

Temperature monitoring, RS485 / Modbus data acquisition and alarm notification for remote equipment and industrial assets - pushed to operators, not discovered on the next site visit.

Environmental Monitoring

Weather stations, air quality, soil and rainfall sensing for agriculture, smart cities and ecological monitoring - solar-powered and wireless options for off-grid sites.

Where customers start their search

Tank Level Monitoring

Continuous remote level readings for tanks and reservoirs.

Pump Monitoring

Run status, faults and alarms for remote pumps.

Solar Site Monitoring

Production and asset data from distributed solar sites.

Remote Meter Reading

Automated meter reads over RS485 / wireless.

Equipment Alarm Notification

Temperature and status alarms pushed in real time.

Temperature & Humidity Controller

Switchgear and cabinet temperature & humidity controller that drives heaters and fans to prevent condensation, with RS485 Modbus for remote monitoring - DIN-rail and panel-mount versions.



Measurement	Temperature + humidity (multi-channel models)
Control	Heater + fan outputs (condensation / anti-dew)
Communication	RS485 (Modbus RTU)
Display	LED, setpoint & alarm
Power supply	AC/DC wide range (model dependent)
Application	Switchgear, ring main units, cabinets
Mounting	DIN-rail / panel-mount

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Protocol Gateway (RS485 <-> Ethernet)

Industrial DIN-rail protocol gateway bridging RS485 (Modbus) field devices to Ethernet / BACnet - for building, facility and equipment monitoring.

**OS**

Linux

Serial

2× RS485

Ethernet

1× 10/100M

Protocols

Modbus RTU/TCP, BACnet

Power supply

AC/DC 12-24V

Mounting

35mm DIN-rail

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Modular Remote I/O System

Expandable DIN-rail remote I/O with PROFINET, EtherCAT and Modbus TCP couplers - for distributed data acquisition and control across facilities.

**Couplers**

PROFINET / EtherCAT / Modbus TCP

Modules

DI / DO / AI / AO, expandable

Ports

2× RJ45 + USB-C config

Power supply

DC 24V

Mounting

35mm DIN-rail

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Connectivity	4G / WiFi / GPS / Ethernet (WAN+LAN)
Serial	RS485 / RS232
Interfaces	USB, Type-C, HDMI
OS	Linux (edge computing)
Power supply	DC 12-24V
Mounting	DIN-rail

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Wireless Communication Gateway

DIN-rail wireless data terminal that collects RS485 / Modbus meter data and uploads it over 4G / LoRa to the cloud - the uplink for distributed energy and equipment monitoring.



Function	RS485 data acquisition + wireless uplink
Wireless	4G / LoRa (model dependent)
Serial	RS485 (Modbus RTU)
Indicators	Link / signal (RSSI) / comm LEDs
Power supply	Wide-range AC/DC
Mounting	35mm DIN-rail

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Ultrasonic Water Level Sensor

Non-contact ultrasonic level measurement for tanks, reservoirs and open channels - for remote water level monitoring.



Measurement	Ultrasonic (non-contact)
Range	0-15 m (model dependent)
Output	4-20mA / RS485 (Modbus)
Power supply	DC 24V
Display	LCD
Protection	IP65
Mounting	Top-mount

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LoRa Temperature & Humidity Sensor

Wireless LoRa temperature and humidity monitoring for rooms, cabinets and remote sites - with local LCD and long-range transmission.



Measurement	Temperature + Humidity
Communication	LoRa (wireless)
Range	Long-range, km-level (line of sight)
Display	Local LCD
Power supply	Battery / DC
Mounting	Wall-mount

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Solar-Powered Weather Station

Self-powered environmental monitoring station with ultrasonic weather sensor and solar panel - wind, rain, temperature, humidity and light for remote, solar and agricultural sites.



Parameters	Wind, rain, temperature, humidity, light
Sensor	Ultrasonic weather sensor
Power	Solar panel + battery (off-grid)
Communication	RS485 / LoRa / 4G (model dependent)
Enclosure	Outdoor, IP65
Mounting	Pole-mount

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Water Quality pH Sensor

Industrial pH electrode with RS485 / 4-20mA output for continuous water-quality monitoring in tanks, treatment plants and process water.



Parameter	pH (ORP / EC variants available)
Range	pH 0-14
Output	RS485 (Modbus) / 4-20mA
Cable	Industrial sealed cable
Application	Tanks, treatment, process water

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Vibration Sensor

Industrial vibration sensor with 4-20mA / RS485 output for pump, motor and rotating-equipment condition monitoring and alarms.

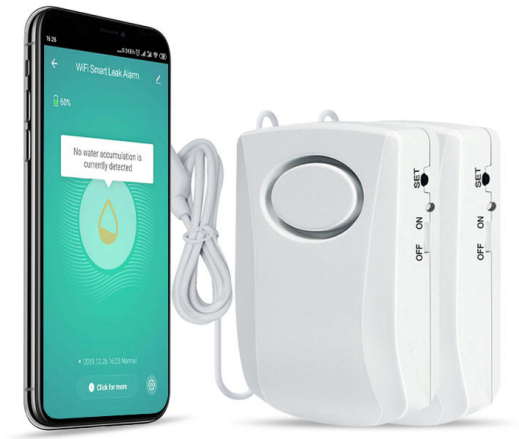


Measurement	Vibration (velocity / acceleration)
Output	4-20mA / RS485 (Modbus)
Application	Pumps, motors, fans, rotating equipment
Function	Condition monitoring + alarm
Mounting	Stud / magnetic base
Enclosure	Industrial metal

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Water Leak Detection Alarm

Wireless water leak detection alarm with probe and app notification - instant alerts for leaks under tanks, pumps and equipment.



Detection	Water leak (probe / cable)
Alert	Local siren + app push
Communication	WiFi (app)
Power	Battery
Settings	On/off, sensitivity
Application	Pump rooms, server rooms, basements

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Ultrasonic Weather Station

All-in-one ultrasonic weather sensor measuring wind speed, wind direction, temperature, humidity and air pressure - no moving parts, with RS485 Modbus output.

**Parameters**

Wind speed & direction, temperature, humidity, pressure (optional light / rainfall)

Wind speed

0-60 m/s

Wind direction

0-360°

Measurement

Ultrasonic, no moving parts

Output

RS485 (Modbus RTU)

Power supply

12-24 VDC

Protection

IP65, pole mount

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Wind Speed & Direction Sensor

Combined anemometer and wind vane for accurate wind speed and direction, with RS485 / 4-20 mA / pulse output options.



Wind speed range 0-60 m/s

Wind speed accuracy ± 0.3 m/s

Wind direction 0-360°, 16 directions

Output RS485 (Modbus) / 4-20 mA / voltage / pulse

Power supply 5-24 VDC

Material Aluminium alloy, IP65

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Tipping-Bucket Rain Gauge

Tipping-bucket rainfall sensor with 0.1 mm resolution and RS485 / pulse output for weather and hydrology stations.



Resolution	0.1 mm
Measurement	Tipping bucket
Max intensity	? 4 mm/min
Caliber	dia. 200 mm
Output	RS485 (Modbus) / pulse
Power supply	5-24 VDC

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Pyranometer (Solar Radiation Sensor)

PV-grade pyranometer measuring total solar radiation (0-2000 W/m²) for solar plants, agriculture and meteorology.



Range	0-2000 W/m ²
Spectral range	300-3000 nm
Accuracy	±5%
Output	RS485 (Modbus) / 4-20 mA / 0-5 V
Power supply	5-24 VDC
Protection	IP65

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Soil Moisture, Temperature & EC Sensor

Buried 3-in-1 soil probe measuring volumetric moisture, temperature and electrical conductivity (EC) with RS485 Modbus output.



Parameters	Moisture, temperature, EC
Moisture	0-100%, $\pm 2-3\%$
Temperature	-40-80 °C
EC range	0-20000 $\mu\text{S}/\text{cm}$
Output	RS485 (Modbus)
Power supply	5-24 VDC
Probe	Stainless steel, IP68

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Air Quality Sensor

Outdoor multi-parameter air quality sensor monitoring PM2.5/PM10, CO2, temperature and humidity, with RS485 Modbus output.



Parameters	PM2.5 / PM10, CO2, temperature, humidity (optional CO / SO2 / NO2 / O3 / noise)
PM range	0-1000 µg/m³
CO2 range	0-5000 ppm
Output	RS485 (Modbus)
Power supply	12-24 VDC
Housing	Louvered, IP65, wall / pole mount

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Dissolved Oxygen Sensor

Online fluorescence (optical) dissolved-oxygen sensor for aquaculture, wastewater and surface water - RS485 Modbus, no membrane or electrolyte.



Method	Fluorescence (optical)
DO range	0-20 mg/L
Accuracy	±0.3 mg/L
Temperature	0-60 °C
Output	RS485 (Modbus) / 4-20 mA
Power supply	9-24 VDC
Protection	IP68

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Turbidity Sensor

Online optical turbidity sensor (90° scattered light) for drinking water, wastewater and environmental monitoring - RS485 Modbus.



Range	0-1000 NTU (optional 0-4000)
Method	90° scattered light
Accuracy	±5%
Output	RS485 (Modbus) / 4-20 mA
Power supply	9-24 VDC
Protection	IP68, immersion

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Water Conductivity (EC) Sensor

Online conductivity / TDS / salinity sensor for water and wastewater monitoring - RS485 Modbus output.



Range	0-20000 $\mu\text{S/cm}$ (optional higher)
Accuracy	$\pm 1.5\%$ FS
Temperature	0-60 $^{\circ}\text{C}$
Output	RS485 (Modbus) / 4-20 mA
Power supply	9-24 VDC
Protection	IP68

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Radar Water Level Sensor

Non-contact radar level sensor for rivers, reservoirs, channels and tanks - unaffected by foam, vapour or fouling, with RS485 Modbus output.



Method	Non-contact radar (FMCW)
Range	0-30 m (model dependent)
Accuracy	±3 mm
Output	RS485 (Modbus) / 4-20 mA
Power supply	12-24 VDC
Protection	IP67

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Atmospheric Multi-Parameter Sensor

Compact radiation-shielded sensor measuring air temperature, humidity, barometric pressure, illuminance and CO2 - RS485 Modbus output.



Parameters	Temperature, humidity, pressure, illuminance, CO2
Temperature	-40-80 °C
Pressure	300-1100 hPa
CO2 range	0-5000 ppm
Output	RS485 (Modbus)
Power supply	12-24 VDC
Housing	Radiation shield, IP65

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Leaf Wetness Sensor

Surface leaf-wetness sensor for crop disease forecasting and smart irrigation in greenhouses and orchards - RS485 Modbus output.



Measurement	Leaf surface wetness / dew
Output range	0-100% wetness
Temperature	-40-80 °C
Output	RS485 (Modbus)
Power supply	12-24 VDC
Protection	IP65

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Split-Core Current Sensor

Clip-on split-core current sensor for non-invasive energy and load monitoring - install without disconnecting existing wiring.



Type	Split-core (clip-on)
Input	AC/DC up to 50A (model dependent)
Output	4-20mA / 0-5V / RS485 (model dependent)
Accuracy	0.5%
Installation	Clip-on, no rewiring
Mounting	Panel / DIN

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Electromagnetic Flow Meter

Flanged electromagnetic flow meter with RS485 / 4-20mA output for water and conductive-liquid flow measurement in pipelines.



Measurement	Electromagnetic
Medium	Water / conductive liquids
Output	4-20mA / RS485 (Modbus)
Connection	Flanged (DN model dependent)
Display	Local LCD
Protection	IP65 / IP68

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Three-Phase Smart Energy Meter

Three-phase four-wire kWh energy meter with RS485 (Modbus) for remote energy data collection and sub-metering across facilities.



Type	Three-phase four-wire
Voltage	3×230/400V
Current	3×10(100)A
Communication	RS485 (Modbus)
Standards	IEC 62052 / 62053
Display	LCD, Class 1

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Multifunction Power Meter

Panel-mount three-phase multifunction power meter measuring voltage, current, power, power factor, frequency and energy, with RS485 Modbus for remote energy monitoring and sub-metering.



Measurement	V, A, kW, kvar, PF, Hz, kWh (three-phase)
Accuracy	Class 0.5S (energy)
Communication	RS485 (Modbus RTU)
Inputs	Via CT / direct (model dependent)
Display	Backlit LCD
I/O	Optional DI / DO / 4-20mA
Mounting	Panel-mount (72 / 96 / 120 mm)

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Wireless IoT Energy Meter

DIN-rail three-phase energy meter with built-in wireless (4G / LoRa / WiFi) for direct cloud upload - no separate gateway needed, ideal for distributed sub-metering and remote sites.



Measurement	V, A, kW, PF, kWh (three-phase)
Wireless	4G / LoRa / WiFi (model dependent)
Wired	RS485 (Modbus) optional
Cloud	Direct platform upload, no gateway
Input	Direct or via CT
Display	LCD
Mounting	35mm DIN-rail

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DIN-Rail Three-Phase Energy Meter

Compact DIN-rail three-phase multifunction energy meter with RS485 Modbus - for panel sub-metering in solar, EV-charging, building and industrial energy management.



Type	Three-phase multifunction, DIN-rail
Measurement	V, A, kW, kvar, PF, Hz, kWh
Accuracy	Class 0.5S (energy)
Communication	RS485 (Modbus RTU)
Input	Direct 80A / via CT (model dependent)
Display	Backlit LCD
Mounting	35mm DIN-rail (7 modules)

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Electrical Fire Monitor (Residual Current)

DIN-rail electrical fire monitoring unit that continuously watches residual (leakage) current and circuit temperature, alarming on faults - with RS485 Modbus for remote electrical-safety monitoring of distribution boxes and cabinets.



Monitoring	Residual current + temperature (multi-channel)
Function	Leakage / over-temperature alarm, electrical fire prevention
Communication	RS485 (Modbus RTU)
Display	LCD + status indicators
Alarm	Audible + relay output
Application	Distribution boxes, cabinets, switchgear
Mounting	35mm DIN-rail

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Why partner with SunGene

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Let's talk

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